

SCENARIO

A company receives **daily employee attendance data** from multiple branch offices.

Every branch sends a CSV file every evening.

Example:

Branch_A_2026-09-07.csv

Branch_B_2026-09-07.csv

Branch_C_2026-09-07.csv

Each file contains:

EmployeeID

EmployeeName

Department

AttendanceDate

CheckInTime

CheckOutTime

Status

The company wants to load the cleaned data into a **SQL Server database**.

The pipeline should run **every day at 11:00 PM**.

1. SOURCE

Daily CSV files from branch offices

Branch A CSV

Branch B CSV

Branch C CSV

2. TARGET

SQL Server



EmployeeAttendance table

3. TRANSFORMATIONS

- Transformation 1 — Date format
- Transformation 2 — Time format

- **Transformation 3 — Department standardization**
- **Transformation 4 — Status standardization**
- **Transformation 5 — Calculate working hours**

4. VALIDATIONS

- EmployeeID must not be NULL
- AttendanceDate must be a valid date
- Department must not be empty
- Status should be one of: present, absent, leave
- WorkingHours ≥ 0
- An employee should not have two identical attendance records for the same date.

5. FAILURE POINTS

- **Failure 1 — File doesn't arrive**
- **Failure 2 — File is corrupted**
- **Failure 3 — Schema changed**
- **Failure 4 — Invalid data**
- **Failure 5 — Database unavailable**
- **Failure 6 — Pipeline stops halfway**

